COIMBATORE INSTITUTE OF TECHNOLOGY

(GOVERNMENT AIDED AUTONOMOUS INSTITUTION)

CIVIL AERODROME POST, COIMBATORE-641014.

PLACEMENT CELL

PLACEMENT YEAR 2024 - 2025

NAME	: B. Rishikessh Balaji
ROLL NUMBER	: 71762103032
DEPARTMENT	: Electrical and Electronics Engineering
EMAIL ID	: rishikesshbalaji2004@gmail.com
CONTACT NUMBER	: 7708510819
COMPANY NAME	: ZOHO
COMPANY TYPE	: Non Dream
JOB DESIGNATION	: Software Developer
SALARY (CTC)	: 7,00,000
INTERN OFFERED	: YES
BOND	: NO

COMMENTS ON SELECTION PROCESS :

ROUND 1 :

Platform: Offline

Round details: Written Test (1 hour)

Areas covered: General Aptitude & C Programming

Your Comments:

Have you cleared the round :

Details about Questions on 1st round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Aptitude	Basic Aptitude questions.	-	-
Technical aptitude	C programming questions, including concepts of arrays and pointers and functions	-	-

ROUND 2 :

Platform: Offline

Round details: It was a simple programming round that focused primarily on developing logical solutions for problems. We were allowed to use any IDE (incl. online ones) and any language of our choice.

Areas covered: Arrays, String Manipulation, Binary Search.

Your Comments: Basic Understanding of language of choice and some logical solving techniques will easily help you get through this.

Have you cleared the round : Yes

Details about Questions on 2nd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Programming	Sprial Matrix Traversal. Basically traverse through a matrix in a spiral manner.	Brute Forced it. Basically traverse in spiral manner and decrease the bounds every cycle.	https://leetcode.com/problems/spiral-matrix/description
Programming	Expression Evaluation. Evaluate an expression represented by a String. For simplicity, you can assume only binary operations allowed are +, -, *, and /. (No paranthesis)	First utilized a stack data type from Java, but the reviewer asked me not to utilize a stack data type and to use an array, so I implemented it with an array but the logic was same.	https://www.geeksforgeeks.org/expression-evaluation/
Programming	Given an array of distinct elements which was initially sorted. This array may be rotated at some unknown point. The task is to find the minimum element in the given sorted and rotated array. It must of complexity O(log(n))	Utilized a modified version of binary search, where I look for the place the rotation took place.	https://leetcode.com/problems/find-minimum-in-rotated-sorted-array/description/
Programming	Given an input string s and a pattern p, implement regular expression matching with support for '.' and '*' where: '.' Matches any single character. '*' Matches zero or more of the preceding element. The matching should cover the entire input string (not partial).	I traversed through both strings and whenever I got a '*' or a '.', I had some If else cases that will take care of the necessary logics. (My code was incomplete but almost there)	https://leetcode.com/problems/regular-expression-matching/description/

ROUND 3 :

Platform: In person (offline)

Round details: Another programming round juts like the one before.

Areas covered: OOPS concepts.

Your Comments: Understanding the applications of OOPS concepts and knowledge of knowing when to implement what will get u through this round.

Have you cleared the round : Yes

Details about Questions on 3rd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
OOPS concept	Build a library Management system, Where using a menu like system, you have to implement the following functions, Register/ Cancel Memberships for library Search for books in the library shelf Check out book for the registered users Return book from the registered users Also implement a fine system, that costed Rs 2 per every day beyond 15 days from date of checking out.	I implemented all of these functions, but honestly, they were focussed more on the way I utilized the OOPS concepts and I was not able to convince them with my methods, they expected better structured code that would flawlessy run in any circumstance. I would personally recommend, asking professors and students from IT based departments about better structured code.	-

ROUND 4 : No round 4

Platform: In person (offline)

Round details: Technical/ HR Interview

Areas covered: Areas of my Interest, Everything that was mentioned in my resume.

Your Comments: I personally was not able to answer most of their questions that concerned IT field, but I answered everything that I had done in my free-time coding projects and interests.

Have you cleared the round : Yes

Details about Questions on 4th round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Programming Logic.	Given n people standing in a circle where 1st has a sword, find the luckiest person in the circle, if, from 1st soldier who is has a sword each has to kill the next soldier and handover the sword to next soldier,	I didn't completely answer the question but the answer is simply to use a logarithmic based	https://www.geeksforgeeks.org/puzzle-100-people-in-a-circle-with-gun-puzzle/

	in turn, the soldier will kill the adjacent soldier and handover the sword to next soldier such that one soldier remains in this war who is not killed by anyone.	solution.	This is not the exact question but also you don't need a Linked List.

AREAS TO PREPARE : Basic Understanding of Programming Logic, Arrays, String Manipulation, OOPS concepts, Some DSA concepts.

SITES / BOOKS YOU SUGGEST FOR PREPARATION FOR THE PROCESS :
https://www.geeksforgeeks.org/
https://leetcode.com/
Aptitude websites (There are many of those)

OVERALL EXPERIENCE: Good, the hiring staff were nice to me, even though I was not upto par of their level and overall it was a pleasant experience throughout the process.

GENERAL TIPS: Even though I didn't know most answers I was answering with confidence and when I didn't know something I basically admitted it and I think that's what they preferred from me, so stay bold.

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PLACEMENT CELL

PLACEMENT YEAR 2024 - 2025

NAME : SHRIRAM R

ROLL NUMBER : 2033033

DEPARTMENT : MSc Decision and Computing Sciences

EMAIL ID : shriramramachandran24@gmail.com

CONTACT NUMBER : 9677751871

COMPANY NAME : ZOHO

COMPANY TYPE : NON - DREAM

JOB DESIGNATION : Software Developer

SALARY (CTC) : 20,000 (stipend)

INTERN OFFERED ? : YES

BOND : NO

*(Please comment this section **in detail** to guide your juniors in a bright way)*

COMMENTS ON SELECTION PROCESS :

ROUND 1 :

Platform: Offline

Round details: General and Technical Aptitude

Areas covered: C Programming, Basic Math

Your Comments: Strong understanding in C basics and pointers, basic aptitude and problem solving skills will be sufficient

Have you cleared the round : YES

Details about Questions on 1st round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)

ROUND 2 :

Platform: Any IDE of out choice (Offline)

Round details: Problem Solving

Areas covered: Arrays, Strings, Stack, Recursion, Binary Search, Dynamic Programming

Your Comments: Regularly practice on LeetCode, aiming to solve medium-level problems quickly. Build a strong base in data structures and algorithms, and understand common problem-solving patterns.

Have you cleared the round : YES

Details about Questions on 2nd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Matrix	Spiral Traversal of Matrix	Brute Force	https://leetcode.com/problems/spiral-matrix/
Array, Stack	Evaluate a mathematical expression following the BODMAS rules Example Input: “3 + 2 * 5” Output: 13	Initially solved using two for loops. Then optimized it work using a single pass	https://leetcode.com/problems/basic-calculator-ii/
Binary Search	Find minimum element in Rotated sorted array of numbers Example Input: [5, 6, 8, 2, 3] Output: 2	Binary Search Approach	https://leetcode.com/problems/find-minimum-in-rotated-sorted-array/
Recursion, Dynamic Programming	Regular Expression Matching Example Input: S = “abbbcc” T = “ab*c.” Output: True	Initially approached using Recusion then added Dynamic Programing to optimize the solution	https://leetcode.com/problems/regular-expression-matching/
String, Stack, Recursion	Decode and expand the given string Example Input: “2[d2[ef]]” Output: defefdefef	Attempted to solve using stack but wasn’t able to pass all test cases	https://leetcode.com/problems/decode-string/
Recursion, Dynamic Programming	Similar to below question: Given an amount and a number of total notes combinations of 100, 200, and 500 rupee notes that equal the given amount. Example Input: N = 1000 Output: 10 The following are possible combinations: [500, 500]		https://leetcode.com/problems/coin-change-ii/

	[500, 200, 200, 100] **[500, 200, 100, 100, 100]** **[500, 100, 100, 100, 100, 100]** **…..**		

ROUND 3 :

Platform: Any IDE of choice (offline)

Round details: Design Round

Areas covered: OOPS

Your Comments: Master the fundamentals of OOP in your preferred language and follow best practices when coding.

Have you cleared the round : YES

Details about Questions on 3rd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Low Level Design	**Develop a Library Management System that enables member registration, book searching, checkout, check-in, and fine calculation.**	**Designed the application's class structure and relations on paper, reviewed it with a proctor, and then implemented the code.**	

ROUND 4 : No round 4

Platform: Offline

Round details: Technical Interview 1

Areas covered: Web Development, Computer Networks, Operating Systems

Your Comments: Had a discussion on projects during the interview. Possess a deep understanding of all technologies listed on my resume, including the trade-offs associated with different approaches and tools.

Have you cleared the round : YES

Details about Questions on 4th round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Web Applications	**How a backend server works ? What is a web application ?**		
Project Discussion	**Discussion on internship project. Difference between TCP and UDP ? Why UDP was used in the project over TCP ?**		
Operating Systems	**Difference between multithreading and multiprocessing. What is physical memory and virtual memory ?**		

ROUND 5 : No round 5

Platform: Offline

Round details: Technical Interview 2

Areas covered: Web Development, Computer Networks, C++, Java, Python, Flask

Your Comments: Have a strong understanding of the strengths, weaknesses, and inner workings of all technologies listed on the resume.

Have you cleared the round : YES

Details about Questions on 4th round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Web Applications	**Define Web application, desktop application and mobile application each in one line. What is frontend ? Give differences between flask and fastApi. What is a framework ? How is it different from a library ? What are REST principles ?**		
Programming Languages	**Explain the use-cases of languages like C, C++ vs Python. Explain the mode of execution for C++, Java and Python. List vs ArrayList in Java. Implement operator overloading in C++ using**		

	pen and paper. Does dynamic allocated memory in C get deallocated automatically when function scope ends ?		
General	Choose a technical topic/concept you like the most and explain why.		

AREAS TO PREPARE :

1. **Intermediate level aptitude**
2. **Strong in preferred OOP based programming language**
3. **Data Structures and Algorithms (Arrays, String, Recurstion, Stack, Dynamic Programming)**
4. **Computer Networks, OS, DBMS**

SITES / BOOKS YOU SUGGEST FOR PREPARATION FOR THE PROCESS :

https://leetcode.com/problemset/all/
https://neetcode.io/practice
https://www.geeksforgeeks.org/
https://takeuforward.org/

OVERALL EXPERIENCE : Good

GENERAL TIPS :

- **During coding rounds make sure to convey your though process to the proctors**
- **Have a deep understanding of the technologies used on resume and their trade offs**

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PLACEMENT CELL

PLACEMENT YEAR 2024 - 2025

NAME : AISHWARYA A

ROLL NUMBER : 71762351003

DEPARTMENT : MCA

EMAIL ID : aishusri182@gmail.com

CONTACT NUMBER : 7904560396

COMPANY NAME : ZOHO

COMPANY TYPE : NON - DREAM

JOB DESIGNATION : Software Developer

SALARY (CTC) :

INTERN OFFERED ? : Intern + PBC

BOND : No

*(Please comment this section **in detail** to guide your juniors in a bright way)*

COMMENTS ON SELECTION PROCESS :

ROUND 1 :

Platform: Written Test

Round details: Quantitative aptitude and C programming (predict output as fillups)

Areas covered: Time, speed and distance, time and work, probability, puzzle

Pointers, basic loops, recursion, shift operators, bit wise operator, Compilation error.

Questions were from easy to medium level

Your Comments: Keep on practicing Aptitude. Understand the pointers concept depth in C.

Have you cleared the round : Yes

ROUND 2 :

Platform: Any IDE

Round details: Short Coding

Areas covered: Dynamic Programming, Stack, Patterns

Your Comments: They have given 6 questions to solve. Out of which I solved 5 questions. The questions were medium to hard. There will be a mentor. On completing each question , they will verify the solution for optimality. Explain your approach more clearly with confidence. I have solved one question in brute force approach if in case we can not find any optimized solution

Have you cleared the round : Yes

Details about Questions on 2nd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Array	**Given an array of positive integers unsorted and distinct numbers with the target k. To find the count of the pairs that add upto k.**	**I solved it using two for loops to iterate over each pair and count the pairs. Since the numbers are distinct we will get only unique pairs**	
Stack	**Given a string of mathematical expression and evaluate the value. If it is not a correct expression to print as invalid expression**		**https://www.geeksforgeeks.org/expression-evaluation/**

Sorting, Array	**Given an array of intervals intervals where intervals[i] = [$start_i$, end_i], return *the minimum number of intervals you need to remove to make the rest of the intervals non-overlapping*.**	**Array can be sorted according to the first element and the numbers are in the list. If the numbers already exist in the list, then count is increased.**	**https://leetcode.com/problems/non-overlapping-intervals/description/**
Spiral Pattern	**Separate patterns for odd n and even n The pattern should be printed as leaving two line in the next spiral. For even it was like concentric squares of one with star other with dots. For odd same like one spiral with start other with dots.**	**For each iteration I have incremented the four pointers two times and stored the structure in a matrix to make it easier.**	
Dynamic Programming	**Given three types of notes with values 100, 200, 500 and count of each notes available as 7. Given an amount and count the number of possible ways that can be formed to get the total amount. Eg:Amount = 1000 n = 5, Possibilities = 7**	**Brute Force: Formed 3 loops Checked for all combinations if it is equal to the amount count is increased.**	
Stack	**Length of the longest Valid paranthesis**		**https://www.geeksforgeeks.org/length-of-the-longest-valid-substring/**

ROUND 3 :

Platform: Any IDE

Round details: Long coding

Areas covered: Library Management System

Your Comments: List, HashMap

Have you cleared the round : Yes

Details about Questions on 3rd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Library Management System	**Given around 6 tasks to do which may also have some sub tasks.**	**I solved only two to three tasks using list and hash map to**	

	Library has some books and their copies. Members can take books available else can do reserations Add book, delete book, add member, delete member, borrow and return book, fine for due dates, limits to take a book for specific period, revenue of the library, reservation, allocation. Generate report available books, members, fine amount, books borrowed by specific member, etc.,	**store the books list, members list and borrowed books list.**	

ROUND 4 : No round 4

Platform: Zoho meet

Round details: Technical round

Areas covered: Puzzles, Concepts in java

Your Comments: Answers can be related to applications of questions rather than giving definitions.

As I was holding two maths degrees already, they asked questions related to applied mathematics. I was able to answer one of the two questions.

Have you cleared the round : Yes

Details about Questions on 4th round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Java	**What is constructor, difference between constructor and methods in a class. Copy constructor**		
Puzzle	**Sand clock puzzle: one clock will measure 4 hrs other can measure 7 hrs. Find a way to calculate 9 hrs**	**I have tried many combinations but cannot solve.**	

AREAS TO PREPARE :

1) Aptitude
2) Problem Solving
3) OOPS (in your preferred language)
4) DSA
5) Core Concepts

SITES / BOOKS YOU SUGGEST FOR PREPARATION FOR THE PROCESS :

1) Geeks for Geeks for core concepts
2) Take you Forward You tube channel for DSA
3) Indiabix for Aptitude

OVERALL EXPERIENCE :

Zoho interview experience was very nice. I was able to communicate with the HRs easily and at each level they will collect the rough sheets we used as to know how we are approach the problem, it is not required to solve it for 100%, they also consider our approach, thinking ability. We should be clear with the task or question given, they will clear your doubts then and there.

GENERAL TIPS :

1. Practice aptitude, coding, dsa regularly.
2. Though we don't know the way to solve try all brute force, and try till end. Don't finish the round till they ask to move, this will make them understand we are trying something either giving up the try.

ALL THE VERY BEST!!

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PLACEMENT CELL

PLACEMENT YEAR 2024 - 2025

NAME : Jayasurya C

ROLL NUMBER : 71762107017

DEPARTMENT : IT

EMAIL ID : cjayasurya5@gmail.com

CONTACT NUMBER : 9345404090

COMPANY NAME : ZOHO

COMPANY TYPE : NON - DREAM

JOB DESIGNATION : Software Developer

SALARY (CTC) :

INTERN OFFERED ? : Yes

BOND : No

*(Please comment this section **in detail** to guide your juniors in a bright way)*

COMMENTS ON SELECTION PROCESS :

ROUND 1 :

Platform: Written Test

Round details: Quantitative aptitude and C programming(predict output)

Areas covered: Standard topics in aptitude. Pointers in C programming.

Your Comments: Keep on practicing Aptitude. Understand the pointers concept depth in C.

Have you cleared the round : Yes

ROUND 2 :

Platform: Any IDE

Round details: Short Coding

Areas covered: Problem Solving

Your Comments: They have given 6 questions to solve. Out of which I solved 4 questions. The questions were medium to hard. There will be a mentor. On completing each question , they will verify the solution for optimality. Explain your approach more clearly with confidence.

Have you cleared the round : Yes

Details about Questions on 2nd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Array	**Given a strictly increasing array arr of positive integers forming a sequence, return *the length of the longest Fibonacci-like subsequence of* arr. If one does not exist, return 0.**	**I solved it using nested for loop.**	**https://leetcode.com/problems/length-of-longest-fibonacci-subsequence/description/**
Stack	**Given an encoded string, return its decoded string.**	**Push all characters to stack. When the character is "]" start popping**	**https://leetcode.com/problems/decode-string/description/**
Sorting, Array	**Given an array of intervals intervals where intervals[i] = [$start_i$, end_i], return *the minimum number of intervals you need to remove to make the rest of the intervals non-overlapping*.**	**I sorted the array based on the end time and then compared and removed the overlapping**	**https://leetcode.com/problems/non-overlapping-intervals/description/**

		interval.	
Sliding window	**Given an integer n, return *the number of ways you can write* n *as the sum of consecutive positive integers.***	**I used Sliding window**	**https://leetcode.com/problems/consecutive-numbers-sum/description/**

ROUND 3 :

Platform: Any IDE

Round details: Long coding

Areas covered: Bit manipulation

Your Comments: Be strong in bit operators.

Have you cleared the round : Yes

Details about Questions on 3rd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Bit manipulation	**Given an array. Find the maximum number in the array. The bits of the maximum number should be the width of all elements in the array. Compress the array to 32-bit integer array. (You have to place all the elements in the array to a 32-bit number)**	**Using left shift for compressing and right shift for decompressing.**	

ROUND 4 : No round 4

Platform: Zoho meet

Round details: Technical round I

Areas covered: All the skills in my resume and find data structure for scenario based questions.

Your Comments: Apply what you learnt practically.

Have you cleared the round : Yes

Details about Questions on 4th round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Projects	**Asked many questions from my projects. Be thorough in your project.**		
OS	**Multiprogramming and Multi Threading, Semaphores, Virtual memory, Memory segments, etc**		
Computer Networks	**TCP connection, How concurrent server works.**		
DSA	**They have given some scenarios and asked me to find what data structure to use there.**		

ROUND 5 : No round 5

Platform: Zoho meet

Round details: Technical round II

Areas covered: Projects, SQL,DSA, Scenario based questions

Your Comments: Be strong in what you know

Have you cleared the round : Yes

Details about Questions on 4th round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
C	**Implementation of Union , Struct**		
DSA	**Given a file consists of all unique words in English. How to search a word in the file.**	**Using set in C++**	
SQL	**Find the maximum product for particular customer in given table**	**Using group by and aggregate functions**	

AREAS TO PREPARE :

1) Aptitude
2) Problem Solving
3) OOPS (in your preferred language)
4) DSA
5) Core Concepts

SITES / BOOKS YOU SUGGEST FOR PREPARATION FOR THE PROCESS :

1) Geeks for Geeks for core concepts
2) Take you Forward You tube channel for DSA
3) Indiabix for Aptitude

OVERALL EXPERIENCE :

Zoho focuses on skills and they are only interested in how you solve the problem and how optimal your solution is.

GENERAL TIPS :

1) Don’t panic.
2) Be confident and keep trying.

ALL THE VERY BEST!!

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PLACEMENT CELL

PLACEMENT YEAR 2024 - 2025

NAME : A Irfaan Fareed

ROLL NUMBER : 71762107015

DEPARTMENT : B.Tech IT

EMAIL ID : irfaanfareed04@gmail.com

CONTACT NUMBER : 9786725825

COMPANY NAME : ZOHO

COMPANY TYPE : NON - DREAM

JOB DESIGNATION : Software Developer

SALARY (CTC) :

INTERN OFFERED ? : Yes

BOND : No

COMMENTS ON SELECTION PROCESS :

ROUND 1 :

Platform: Written Test

Round details: Quantitative aptitude and C programming (predict output)

Areas covered: Standard topics in aptitude. Pointers in C programming.

Your Comments: Keep on practicing Aptitude. Understand the pointers concept depth in C.

Have you cleared the round : Yes

ROUND 2 :

Platform: Any IDE

Round details: Short Coding

Areas covered: Problem Solving, DSA

Your Comments: This round was not held on any typical coding platforms, We were asked to use any code editors and any programming language of our choice. Initially 2 questions were given, and the output was verified by the invigilator for each question. When solved correctly, further 2 more questions, little advanced than the previous ones were given.

Have you cleared the round : Yes

Details about Questions on 2nd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Matrix	**Given a matrix of dimension m x n, return an array of the matrix elements in spiral order**	**I solved it using pop and iterating through the limits of the spiral.**	**https://takeuforward.org/data-structure/spiral-traversal-of-matrix/**
String Manipulation	**Given a string s which represents an expression, evaluate this expression and return its value.**	**Maintain a dictionary containing operators as keys with their precedence priority as values and split the given expression into pairs separated by operators and follow the precedence**	**https://leetcode.com/problems/basic-calculator-ii/solutions/4556919/very-easy-to-understand-and-aslo-beats-96-84-of-users-with-python3/**
Arrays, Searching Techniques	**Find the number of times a given sorted array is rotated with time complexity of O(log n)**	**Use binary search to return the index of the minimum element.**	**https://www.geeksforgeeks.org/find-rotation-count-**

			rotated-sorted-array/
Sliding window	**Given two strings S and P where S consists of only lowercase English alphabets while P consists of lowercase English alphabets as well as special characters '.' and '*', the task is to implement a function to test regular expression such that:** • **'.' Matches any single character.** • **'*' Matches zero or more occurrences of the preceding element.**	**Create a boolean 2D array that holds 0 or 1 whether if particular character can be present in that position or not.**	**https://www.geeksforgeeks.org/implementing-regular-expression-matching/**

ROUND 3 :

Platform: Any IDE

Round details: Long coding

Areas covered: OOPs

Your Comments: Make yourself comfortable to be able to work with classes and objects and must have clear understanding of the problem.

Have you cleared the round : Yes

Details about Questions on 3rd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Object Oriented Programming	**Implement a library management system, facilitating registration and cancellation of membership, search books through shelf and row (row and column) in the library, lend books, return books, fine implementation for overdue, and book renewal.**	**Create classes for member, book and librarian with respective attributes.**	

ROUND 4 :

Platform: Offline

Round details: Technical + HR round

Areas covered: Projects from resume, OS, DBMS, Networking concepts, OOPs, DSA.

Your Comments: Have a deep insight about your projects and make sure you are strong in whatever domain you say you are comfortable with.

Have you cleared the round : Yes

Details about Questions on 4th round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Self-Introduction	**Tell me about yourself. Enlighten me about your family background.**		
Projects	**Asked many deep questions from my projects and the libraries that I had used.**		
OS	**What is virtual memory and how is it different from physical memory? Suppose you are downloading a game, which parts would occupy the virtual memory and which parts would occupy the physical memory.**	**Told the definition of virtual memory and its difference from physical memory. The game code and inactive assets go to virtual memory, while active game code, assets, and real-time data are in physical memory (RAM).**	
DBMS	**What are locks and why are they used? What is a database and how is advantageous compared to storing data in files?**	**Told the definition of locks and its types (Shared and Exclusive) and also talked about Semaphores and Mutex variables. Gave definition of database. It provides better data integrity, faster querying, efficient data management, concurrent access control, data security, and reduces redundancy.**	
Computer Networks	**What is a web-server? How do 2 devices in a network communicate or send/receive packets?**	**Gave definitions with examples. Told about packets, routers, switches, OSI model and sockets.**	

DSA	**What is a hash map and how can it be implemented (if in python how you will implement a dictionary)?**	**Gave definition of hash map and told the hash function can either be a simple memory offset(addition) or a mod function of the memory address to map the key to the address of the value.**	

AREAS TO PREPARE :

1) Aptitude
2) Problem Solving
3) OOPS (in your preferred language)
4) DSA
5) DBMS
6) Networking

SITES / BOOKS YOU SUGGEST FOR PREPARATION FOR THE PROCESS :

1) Geeks for Geeks for core concepts
2) Striver's Take you Forward for DSA
3) Indiabix for Aptitude

OVERALL EXPERIENCE :

The whole process felt natural not like a typical placement process but like lab practicals and constructive interactions. The interviewer gave a point that "If one needs to create a product-based company, DBMS, Networking, OS and OOPs are the 4 main pillars, you need not know everything but if you are very strong in what you know then it is more than sufficient". The recruiters were very supportive and also gave constructive criticisms and gave us suggestions on how to improve as well.

GENERAL TIPS :

1) Read the question thoroughly before getting started.
2) Don't get fixated on only a certain perspective of a solution to a problem.
3) Practice lots of problems (both aptitude and coding) to improve your speed since time is of the essence.
4) Don't feel downed when you don't get the expected results because the outcome may surprise you as it did happen in my case.

WISHING YOU ALL SUCCESS!!
ALL THE VERY BEST!!

COIMBATORE INSTITUTE OF TECHNOLOGY

(GOVERNMENT AIDED AUTONOMOUS INSTITUTION)

CIVIL AERODROME POST, COIMBATORE-641014.

PLACEMENT CEL

PLACEMENT YEAR 2024 - 2025

NAME : Siranjith Rajendran

ROLL NUMBER : 71762107047

DEPARTMENT : Information Technology

EMAIL ID : siranjithnewit@gmail.com

CONTACT NUMBER : 8438034782

COMPANY NAME : ZOHO

COMPANY TYPE : NON - DREAM

JOB DESIGNATION : Software Developer

SALARY (CTC) :

INTERN OFFERED ? : Yes

BOND : No

*(Please comment this section **in detail** to guide your juniors in a bright way)*

COMMENTS ON SELECTION PROCESS :

ROUND 1:

Platform: Written Test

Round details: Quantitative aptitude and C programming output prediction. MCQ's will not be present

Areas covered: Topics such as Time and Distance, Percentage, SI, Volume, Partnership, etc, are covered in Quantitative aptitude and in C Programming questions from Pointers, Recursion, Keywords and bitwise operators were asked.

Your Comments: Practice Aptitude questions and learn C language (focus on pointers, operators and keywords).

Have you cleared the round: Yes

ROUND 2:

Platform: Any IDE

Round details: Short Coding

Areas covered: Problem Solving

Your Comments: There were six questions. In my group, initially, two questions were given. After solving those questions, we were allowed to receive the next two. The problems ranged from medium to hard. I've written complete code for two questions, one partial implementation and intuition for one question.

Have you cleared the round: Yes

Details about Questions on 2nd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
2D Array	**Given a matrix of dimension m x n, return an array of the matrix elements in spiral order**	**I solved it using nested loop.**	**https://leetcode.com/problems/spiral-matrix/description/**

String manipulation	**Given an string to find the value for the expression. Eg. Input:2+3*5 Output:17**	**I used stack to perform the operations.**	
Sorted Rotated Array	**Given an array of distinct elements which was initially sorted. This array may be rotated at some unknown point. The task is to find the minimum element in the given sorted and rotated array(search should be performed in O(log n) time complexity).**	**I used binary search with some modifications**	**https://www.geeksforgeeks.org/problems/minimum-number-in-a-sorted-rotated-array-1587115620/0**
Pattern Matching (Sliding window)	**Given two strings S and P where S consists of only lowercase English alphabets while P consists of lowercase English alphabets as well as special characters '.' and '*', the task is to implement a function to test regular expression such that: • '.' Matches any single character. • '*' Matches zero or more occurrences of the preceding element.**	**I explained the intuition using sliding window concept**	

ROUND 3:

Platform: Any IDE

Round details: Advanced Coding

Areas covered: Design

Your Comments: Initially, Part-1 and Part-2 of the questions were given. Then, they provided consecutive parts, Part-3, Part-4, and Part-5, after completing each question. The consecutive parts should be added to the previous work without much effort and changes. Modular approach is recommended. A strong grasp of OOPS concepts and some creativity are required.

Have you cleared the round: Yes

Details about Questions on 3rd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Design (Using OOPS)	**To design a library management system With Librarian being able to add members and remove members. Then facilities like book checkout, book renew, return, waiting and fine calculation. Books to be placed in a shelf containing five rows and five columns. Single row can accommodate 5 books.**	**I've used three classes they are Member, Librarian and Reservation. Reservation class inherited Librarian class. Librarian class will store Member objects in an array(Array of objects). Librarian class will perform the operations of adding and removing the members. Reservation class will perform the operations of book checkout, renew, return, reserve the unavailable book, search and fine calculation**	

ROUND 4:

Platform: Offline (FTF)

Round details: Technical and General HR

Areas covered: DSA, C++, DBMS, Project, Internship and Background.

Your Comments: Stay calm and communicate in proper manner. Have some basic understanding on the CS fundamentals. Be strong in one programming language.

Have you cleared the round : Yes

Details about Questions on 4th round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
C++	**Function pointer, Namespace, Class template, OOPs based questions, Diamond Problem, difference between abstract and general(General Knowledge), Call by reference and value difference, advantages and disadvantages of C++, header(why it is used?) and difference between struct and class.**		
DSA	**1)What is datastructure and my we need datastructures?** **2)Asked to list all the datastructures upto my knowledge and explain each datastructure in detail.** **3) What is AVL tree?** **Perform rotations in AVL tree.** **4) Graph traversal questions(based on BFS and DFS)**		
DBMS	**ACID properties, joins(what is joins and its types) and Normalization.**		
Project and Internships	**1)Project based questions and what I learned in the Internship.** **2)A question was raised whether I have faced any problem in building my project? How did I debug and solve the problem**		
General HR questions	**1) Introduce Yourself.** **2) Tell me about your background?** **Questions were asked with respect to my background** **3) Greatest achievement in your life?** **4) Any questions do you want to ask?**		

ROUND 5: No round 5

AREAS TO PREPARE:

1) Aptitude
2) Problem Solving and Enhance your logical thinking.
3) OOPS
4) DSA
5) Computer Networks
6) OS
7) DBMS
8) Be strong in any one programming language (C++ or Java is recommended, python is also okay)

SITES / BOOKS YOU SUGGEST FOR PREPARATION FOR THE PROCESS:

1) GFG and InterviewBit for core concepts
2) Leetcode and Code Ninja for problem solving.
3) Indiabix and Careerride for Aptitude

OVERALL EXPERIENCE:

Zoho cares more about character and way of thinking. They don't expect us to have a ton of achievements and to have perfect knowledge in every concepts. Just knowing the basics and having good problem-solving skills is enough. Communication wasn't an issue either, I was able to speak in Tamil during the entire Technical HR and coding interview, which made the whole process a lot easier for me.

GENERAL TIPS:

1) Don't hesitate or be afraid to communicate with HR personnel.
2) If you are unable to figure out a proper and complete solution to the given problems in coding interview do not get panic. Discuss your approach to the problem with HRs.
3) Be calm, be humble and be yourself.

WISH YOU ALL SUCCESS!

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PLACEMENT CELL

PLACEMENT YEAR 2024 - 2025

NAME : Narmatha R

ROLL NUMBER : 71762105088

DEPARTMENT : CSE

EMAIL ID : narmatha200329@gmail.com

CONTACT NUMBER : 9363390951

COMPANY NAME : ZOHO

COMPANY TYPE : NON - DREAM

JOB DESIGNATION : Software Developer

SALARY (CTC) :

INTERN OFFERED ? : Yes

BOND : No

*(Please comment this section **in detail** to guide your juniors in a bright way)*

COMMENTS ON SELECTION PROCESS :

ROUND 1 :

Platform: Written Test

Round details: Quantitative aptitude and C programming(predict output)

Areas covered: Standard topics in aptitude. Pointers in C programming.

Your Comments: Keep on practicing Aptitude. Understand the pointers concept depth in C.

Have you cleared the round : Yes

ROUND 2 :

Platform: Any IDE

Round details: Short Coding

Areas covered: Problem Solving

Your Comments: They have given 6 questions to solve. Out of which I solved 4 questions. The questions were medium to hard. There will be a mentor. On completing each question , they will verify the solution for optimality. Explain your approach more clearly with confidence.

Have you cleared the round : Yes

Details about Questions on 2nd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Array	**Given a strictly increasing array arr of positive integers forming a sequence, return *the length of the longest Fibonacci-like subsequence of* arr. If one does not exist, return 0.**	**I solved it using nested for loop.**	**https://leetcode.com/problems/length-of-longest-fibonacci-subsequence/description/**
Stack	**Given an encoded string, return its decoded string.**	**Push all characters to stack. When the character is "]" start popping**	**https://leetcode.com/problems/decode-string/description/**
Sorting, Array	**Given an array of intervals intervals where intervals[i] = [$start_i$, end_i], return *the minimum number of intervals you need to remove to make the rest of the intervals non-overlapping*.**	**I sorted the array based on the end time and then compared and removed the overlapping**	**https://leetcode.com/problems/non-overlapping-intervals/description/**

		interval.	
Sliding window	**Given an integer n, return *the number of ways you can write* n *as the sum of consecutive positive integers.***	**I used two for loops. When the subsequence sum exceeds the given number then I changed the pointer.**	**https://leetcode.com/problems/consecutive-numbers-sum/description/**

ROUND 3 :

Platform: Any IDE

Round details: Long coding

Areas covered: Bit manipulation

Your Comments: Be strong in bit operators.

Have you cleared the round : Yes

Details about Questions on 3rdround:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Bit manipulation	**Given an array. Find the maximum number in the array. The bits of the maximum number should be the width of all elements in the array. Compress the array to 32-bit integer array. (You have to place all the elements in the array to a 32-bit number)**	**Using left shift for compressing and right shift for decompressing.**	

ROUND 4 : No round 4

Platform: Zoho meet

Round details: Technical round I

Areas covered: All the skills in my resume and find data structure for scenario based questions.

Your Comments: Apply what you learnt practically.

Have you cleared the round : Yes

Details about Questions on 4th round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Projects	**Asked many questions from my projects. Be thorough in your project.**		
OS	**Multiprogramming and Multi Threading, Semaphores, Virtual memory, Memory segments, etc**		
Computer Networks	**TCP connection, How concurrent server works.**		
DSA	**They have given some scenarios and asked me to find what data structure to use there.**		

ROUND 5 : No round 5

Platform: Zoho meet

Round details: Technical round II

Areas covered: Python, Scenario based questions

Your Comments: Be strong in what you know

Have you cleared the round : Yes

Details about Questions on 4th round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Python	**Asked me many questions from python. And to implement set() in python using time complexity O(1).**		
DSA	**Given a file consists of all unique words in English. How to search a word in the file.**	**Using set in python or Hash set in Java**	

AREAS TO PREPARE :

1) Aptitude
2) Problem Solving
3) OOPS (in your preferred language)
4) DSA
5) Core Concepts

SITES / BOOKS YOU SUGGEST FOR PREPARATION FOR THE PROCESS :

1) Geeks for Geeks for core concepts
2) Take you Forward You tube channel for DSA
3) Indiabix for Aptitude

OVERALL EXPERIENCE :

Zoho focuses on skills, not just specific programming languages like Java, C, or C++. I am a Python programmer, but I cleared all the rounds. It's about your understanding of the language. Make sure to understand it in depth. They asked me not to use any built-in functions. They are only interested in how you solve the problem and how optimal your solution is. Communication isn't a major concern; my second technical round was conducted in Tamil.

GENERAL TIPS :

1) Don’t panic. Stay relaxed and think before solving it.
2) You can achieve anything when the time is right. So be confident and keep trying.

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